

Leif Hellgren

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SUMMARY

Physics-trained engineer who owned a flight-critical subsystem end to end at Blue Origin, from first-principles modeling through hardware delivery and test. Since then I've shipped production software independently: API integrations, LLM-driven automation, and an iOS app I'm building now. I write my own code rather than waiting on someone else's tools.

EXPERIENCE

Responsible Engineer, Flight-Critical Cryogenic Fluid Management System (Lunar Lander) | Blue Origin

June 2022 - November 2024 | El Segundo, CA

- Owned a flight-critical propellant tank pressure management subsystem on Blue Origin's first lunar lander end to end, from concept through design, analysis, procurement, and test readiness, using a thermodynamic venting approach to manage tank pressure in microgravity where traditional venting was not viable.
- Built a first-principles fluid and thermal modeling tool in Python and MATLAB to size components and drive design decisions ahead of formal analysis cycles.
- Designed the restrictor, latch valve, and heater assembly, and owned design of a large cryogenic tubing assembly for LH2 and LOx routed through a constrained vehicle environment, reducing system mass while meeting flow, structural, and thermal requirements.
- Led the subsystem FMECA and an independent hydrogen accumulation and ignition risk analysis for off-nominal conditions, and the mitigations went into the vehicle design.
- Defined test requirements and procedures, designed test rigs, and coordinated with vendors through development, acceptance, and qualification testing.
- Coordinated with thermal, GNC, structures, manufacturing, integration, safety, and test throughout, and owned the subsystem's drawings, releases, BOMs, and P&ID.
- Selected vendors, wrote specs, and managed fabrication and delivery schedules for the subsystem.
- Addressed the subsystem in its entirety at PDR, CDR, and TRR.

Independent Developer, Python & LLM Systems | Self-Employed

November 2024 - April 2026

- Built a Python job-board monitor integrating 3 ATS APIs (Ashby, Greenhouse, Lever) across 27 companies, with tiered relevance classification, deduplication, and scheduled automation.
- Reverse-engineered an undocumented HTML export format to build an automated resume-tailoring pipeline with headless Chrome PDF rendering, fixing an escaping bug that would have silently corrupted output files.
- Built a full software stack for stock market analysis: a Streamlit analytics application over Pandas, a scanner filtering thousands of tickers against defined criteria, and a staged alerting pipeline with webhook delivery.
- Currently building an iOS app that pairs gym and macro tracking with an integrated LLM agent tailored to the app's purpose.

Founder | Aerospace Research Cooperative, UC Santa Barbara

2017-2021

- Founded UCSB's first aerospace organization and collaborated with the university's Experimental Cosmology Lab.

Co-Founder & Lead Propulsion Feed System Engineer | UCSB Rocket Propulsion Laboratory

2017-2021

- Co-founded what became the university's largest engineering organization and an approved senior capstone project, and led the design of a methalox engine feed system.

PROJECTS

Growth Consultant | Launchpoint

June - July 2026

Case study: building the outbound motion and CRM from scratch (read [here](#))

- Shipped a feature into a live production codebase (Next.js/TypeScript/Drizzle ORM) with no ramp time, surfacing media performance metrics in a dashboard modal.

EDUCATION

Bachelor's in Physics | University of California, Santa Barbara

2017-2021

SKILLS

Software & Data: Python (Pandas, NumPy, Streamlit), TypeScript/Next.js, SQL, Drizzle ORM, Git, MATLAB, Pine Script

Engineering: systems engineering, mechanical design (CAD, P&IDs, BOMs), fluid and thermal analysis, first-principles modeling, FMECA and risk analysis